

Philosophical Logic Task B

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(a) List the main differences between first- and second-order logic.

Response: The following are the differences between FOL and SOL

- SOL includes predicate variables while FOL only includes variables whereby the variables are terms
 - Therefore, for any PC-model $\mathcal{M}$, a variable assignment $g \langle \mathcal{D}, \mathcal{I} \rangle$ assigns a member of the domain to variable terms only in FOL, while in SOL, g also assigns an n -place relation over $\mathcal{D}$ to the n -place predicate variable
- SOL allows for quantification over predicates while FOL only allows for quantification over terms
- As a result, SOL has a greater expressive power over FOL
- More wffs are valid in SOL than in FOL

(b) Briefly expound and critically assess what you take to be the best argument for

(i) taking second-order logic to be logic (500 words)

Response: The strongest argument for SOL being logic is Boolos's argument. Boolos's argument mainly seeks to refute Quine's own argument against SOL being considered logic (I will not explicate Quine's argument since I present it in the next question). In particular, he argues that, contra Quine, quantifying over predicates merely means that a predicate F has a range, rather than being a name position. Furthermore, he argues that SOL does not in fact have many ontological commitments – the formula $\exists X \exists x \exists y (Xx \wedge Xy \wedge \sim x = y)$ is not valid in SOL and so SOL is not committed to the existence of a two-membered set. I think his criticisms of Quine's argument goes wrong here. First, it is difficult to see why just because a predicate F has a range means that it is not a name position. Consider the analogy of $\exists x (x \text{ walks})$. Here, we see that the variable term x also has a range (perhaps of people), but this nonetheless does not defeat the fact that it is a name position because we substitute x for names of various people. Second, I think Boolos misattributes the force of Quine's objection to SOL's ontological commitments. It is not the mere fact that SOL has vast ontological commitments to the existences of sets, but rather the mere fact that it is committed to certain types of entities, when logic is supposed to be theory-neutral with as little commitments as possible.

Boolos's reply to the second objection I made is as such: he thinks that notions of set, class, property, concept, and relation have been considered to be distinctively logical notions. This is because one can say that a set itself is topic-neutral – anything can belong to a set, have a property or be related to something. So, if one's necessary criterion for something to be logic is that it is theory neutral, and if Boolos is right, then he seems to have made some progress favouring SOL as logic. But I do not think that this is an effective reply. While anything might be included in a set, such as Socrates or numbers, the problem is that what a set is entails certain kinds of properties i.e. the axiom of choice or the axiom of extensionality, which may not hold for other kinds of entities such as attributes. For instance, the sets of animals who have hearts and animals who have kidneys are the same set by the axiom of extensionality, but are clearly not the same attribute. SOL therefore commits itself to a certain kind of (non-logical) structure and assumptions working in the background. This is why Boolos' reply fails, and why his general argument fails.

(ii) taking second-order logic to not be logic (500 words)

Response: The strongest argument against SOL being logic is Quine's. I reformulate Quine's argument to be the following:

1. SOL embeds set-theory into its conception of 'logic'
2. Set theory is not logic
3. Therefore, SOL is not logic

On what grounds does Quine justify (i)? Quine draws an analogy between ordinary quantification in FOL and name positions – when we say $\forall x(x \text{ walks})$, we treat x as a name position such as 'Matthew'. Likewise, when we quantify over predicates, we treat the predicate as name positions as well for some sort of entity. Logicians then treat predicate letters as sets because they want to find a way to establish the sameness of predicates, which follows from the axiom of extensionality which holds that two sets are equivalent iff their extensions are the same; attributes do not have this property – animals which have hearts might have the same extension as the attribute of animals having kidneys, but they nonetheless aren't the same attribute. And if we treat predicate variables as name positions for sets, then we smuggle in truths of set theory such as $\exists G\forall x(Gx \leftrightarrow Fx)$ as logic validities since this follows from the SOL tautology $\forall x(Fx \leftrightarrow Fx)$.

Just because SOL includes more logical truths than FOL is not ipso facto a reason to deny SOL the status of logic. Quine therefore needs to prove that (ii) is true to reach the conclusion expressed in (iii). I think his argument here is less clear. He argues that "We can in this way enjoy the convenience of an ontology of sets...without footing the ontological bill" but that there will be an "ontological reckoning". It is not clear what this ontological reckoning is. However, I take Quine's argument against Set Theory being logic as a pragmatic argument. Logic is supposed to be neutral of theory – even set theory. Therefore, it must have the lowest ontological commitment possible. Quine therefore argues that we need not use predicate variables to admit sets as "values of quantified variables", and can instead use the extralogical vocabulary of " $x \in y$ " to resolve this need. We can define the membership criteria of a set in FOL without appealing to SOL. Crucially, I think that we can add the fact that to include set theoretical truths as logical validities is to presuppose a commitment to a certain theory (ZFC in this case) and their accompanying axioms which might not be logical tautologies. In short, Set Theory is not logic because logic is supposed to be theory-neutral, free of ontological and theoretical commitments that Set Theory makes. Logic is a theory-neutral language to express our arguments to ensure truth-preservation in virtue of the argument's form and structure. In this sense, because SOL is motivated to help reduce mathematics into logic, it invariably adds extra-logical notions and commitments to FOL, rendering it no longer logic as we conceive of it.